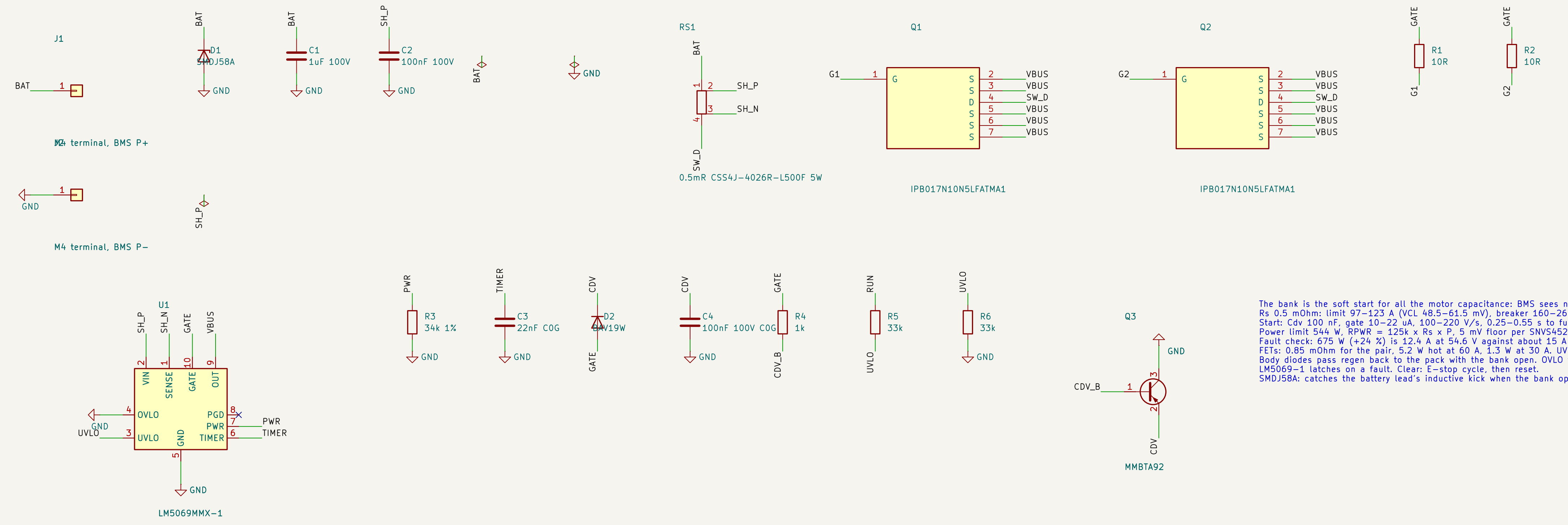
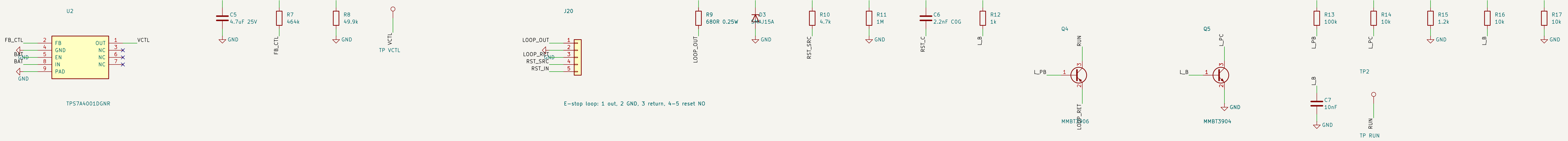
1. Battery in and E-stop switch (LM5069-1, two FETs)


J1
BAT
M4 terminal, BMS P+
GND
M4 terminal, BMS P-
D1 SMDJ58A
C1 1uF 100V
C2 100nF 100V
RS1
SH_P
SH_N
0.5mR CSS4J-4026R-L500F 5W
Q1 IPB017N10N5LFATMA1
Q2 IPB017N10N5LFATMA1
VBUS
SW_D
G1
G2
R1 10R
R2 10R
U1 LM5069MMX-1
UVLO
OVLO
PWR
TIMER
GATE
R3 34k 1%
C3 22nF C0G
D2 15V
C4 100nF 100V C0G
R4 1k
R5 33k
R6 33k
Q3 MMBTA92
CDV_B

The bank is the soft start for all the motor capacitance: BMS sees no inrush. Reset closes it, loop open drops it. No software in the path. Compute is not behind it.
Rs 0.5 mOhm: limit 97–123 A (VCL 48.5–61.5 mV), breaker 160–260 A (VCB 80–130 mV). Above peak draw and JK DOC 80 A. 1.8 W at 60 A.
Start: Cdv 100 nF, gate 10–22 uA, 100–220 V/s, 0.25–0.55 s to full. Bus ceiling 9.4 mF at 2.06 A hot DC SOA line (112 W, 45 K Tj rise). Drive caps unmeasured.
Power limit 544 W, RPWR = 125k x Rs x P, 5 mV floor per SNVS452G sec 9.2.1.2.3. Timer 22 nF: 1.0 ms typ, 1.8 ms max.
Fault check: 675 W (+24 %) is 12.4 A at 54.6 V against about 15 A at 1.8 ms, 70 C, one FET, scaled from the 10 ms curve as sqrt(t). Read the real curve at layout.
FETs: 0.85 mOhm for the pair, 5.2 W hot at 60 A, 1.3 W at 30 A. UVLO low sinks the gate at 2 mA: about 170 us off.
Body diodes pass regen back to the pack with the bank open. OVLO tied low: opening on overvoltage would strand regen on the bus.
LM5069-1 latches on a fault. Clear: E-stop cycle, then reset.
SMDJ58A: catches the battery lead's inductive kick when the bank opens. Clears 54.6 V, FETs and U1 are 100 V.

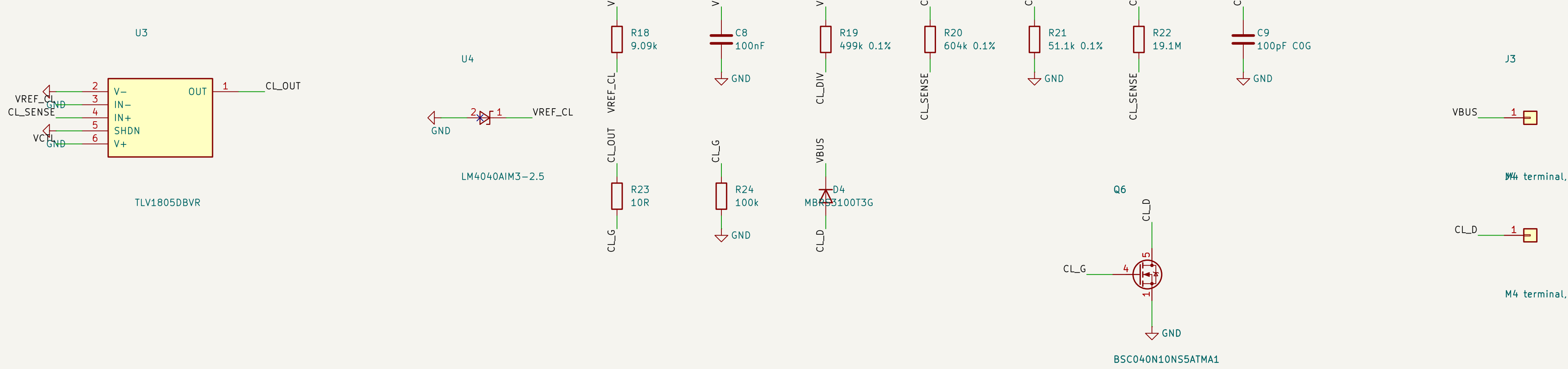
2. Control rail and E-stop latch (no logic IC)


U2 TPS7A4001DGNR
FB_CTL
VCTL
C5 4.7uF 25V
R7 464k
R8 49.9k
TP1
TP VCTL
J20
LOOP_OUT
LOOP_RET
RST_SRC
RST_IN
E-stop loop: 1 out, 2 GND, 3 return, 4–5 reset NO
R9 680R 0.25W
D3 SMAJ15A
R10 4.7k
R11 1M
C6 2.2nF C0G
R12 1k
Q4 MMBT3906
Q5 MMBT3904
L_PB
L_PC
L_B
RUN
C7 10nF
TP2
TP RUN
R13 100k
R14 10k
R15 1.2k
R16 10k
R17 10k

12.1 V, 11 mA. On the battery side, so regen through the body diodes keeps it up with the BMS open. Runs the latch and the clamp. 0.5 W at 54.6 V.

Seal-in latch, ngspice checked at hFE 80: stays off at power-up even with reset held, reset closes it, a 50 us contact bounce rides through,
loop open drops it in about 100 us, re-closing the loop does nothing, a reset held through a loop cycle does nothing, a new press works 5 ms after release.
RUN 7 V (UVLO 3.5 V): needs 5.7–6.1 V rising, so it holds. Loses VCTL, RUN drops, UVLO drops: bank opens.
Loop 6 mA, wetting for a control-rated NC contact. No 12 V on the connector without a series R, pin 1 next to GND: a short there opens the loop.
Brake resistor NC thermostat: wire it in series in the loop harness, no board parts. E-stop model still to confirm.

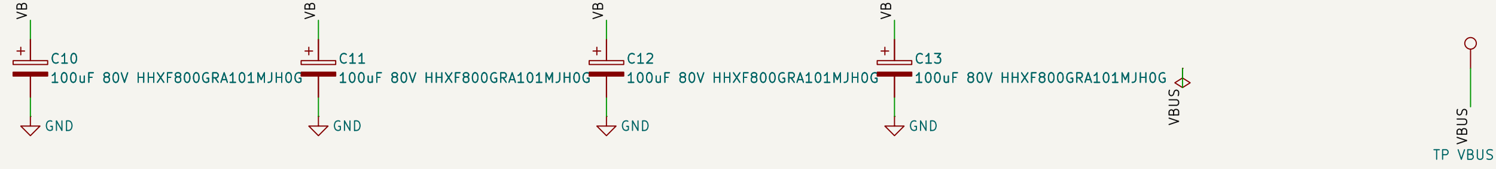
3. Braking clamp: 56.8 V on, 55.8 V off, 57.5 V worst case, into the HS25


U3 TLV1805DBVR
VREF_CL
CL_SENSE
CL_OUT
U4 LM4040AIM3-2.5
R18 9.09k
C8 100nF
R19 499k 0.1%
R20 604k 0.1%
R21 51.1k 0.1%
R22 19.1M
C9 100pF C0G
R23 10R
R24 100k
D4 MBR0100T3G
Q6 BSC040N10NS5ATMA1
CL_G
CL_D
J3
VBUS
M4 terminal, HS25 +
M4 terminal, HS25 −

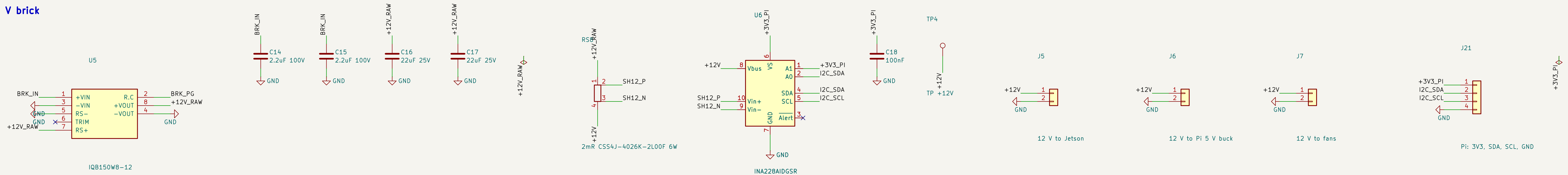
Arcol HS25 10 Ohm on the frame plate: 322 W, 5.7 A. Threshold values are rev A's, unchanged.
Self-powered: VCTL lives as long as the bus or the pack does. If the FET sticks on, the plate thermostat opens the loop and the bank drops the bus.
TLV1805 drives the FET directly: 100 mA peak, 0.3 us into 4 nF. Rev A's driver, meter, shunt, status FET and NTC are gone, and a smaller FET replaces the IPT015.
Leg stage later: fit 2 Ohm and recheck this FET, 29 A gives about 5 W.

4. Bus capacitors

400 uF, 80 V. The drives have their own. Can is 10 x 16.5: land owed, and it only clears under the higher ceiling, away from the fans.

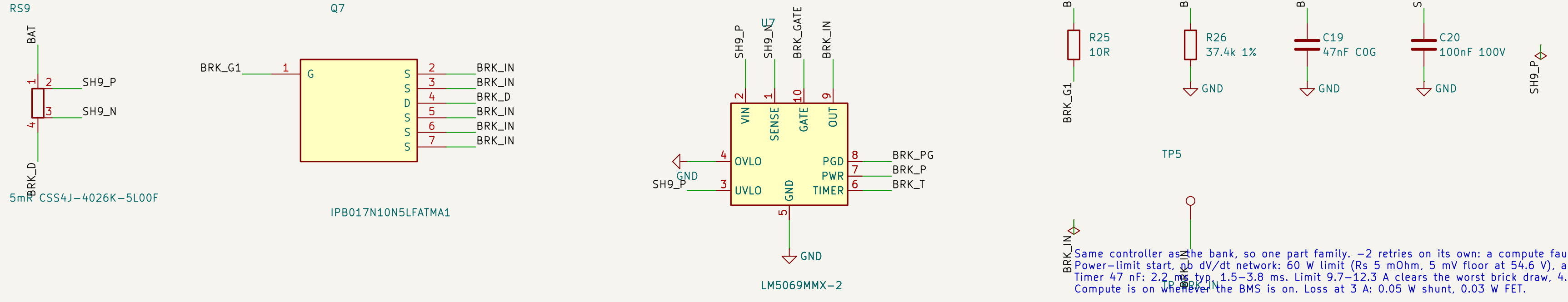

VBUS
C10 100uF 80V HHXF800GRA101MJH0G
C11 100uF 80V HHXF800GRA101MJH0G
C12 100uF 80V HHXF800GRA101MJH0G
C13 100uF 80V HHXF800GRA101MJH0G
GND
TP3
TP VBUS

5. 12 V brick


U5 IQB150W8-12
BRK_IN
BRK_PG
+12V_RAW
+VIN
−VIN
RS−
TRIM
RS+
R.C
+VOUT
−VOUT
GND
C14 2.2uF 100V
C15 2.2uF 100V
C16 22uF 25V
C17 22uF 25V
RS10
SH12_P
SH12_N
2mR CSS4J-4026K-2L00F 6W
U6 INA228AIDGSR
+3V3_PI
I2C_SDA
I2C_SCL
C18 100nF
TP4
TP +12V
J5
12 V to Jetson
J6
12 V to Pi 5 V buck
J7
12 V to fans
J21
Pi: 3V3, SDA, SCL, GND

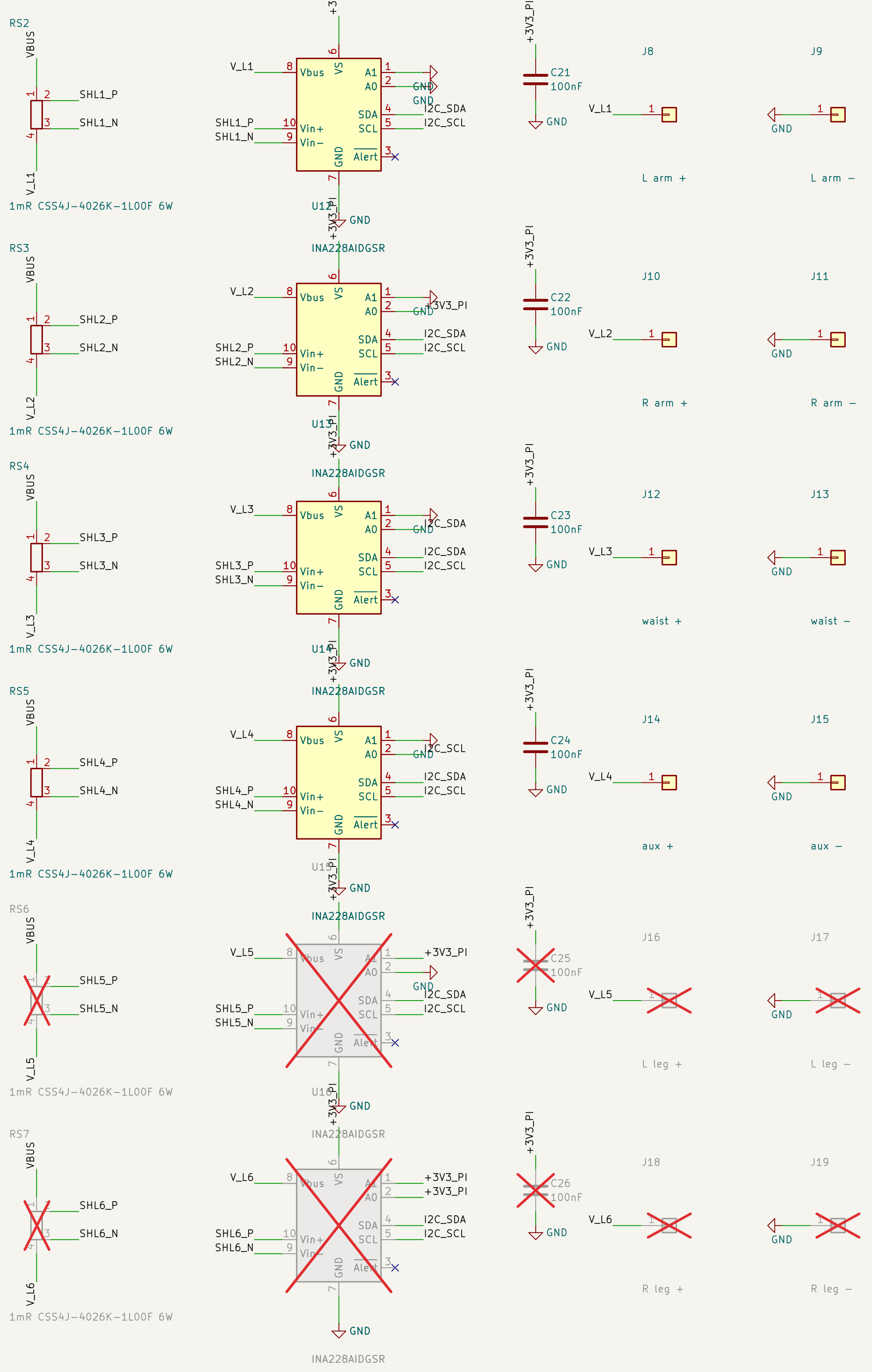
Ahead of the E-stop: motors off, compute stays up. Soft start in block 5a, BRK_PG on R.C keeps the brick off until its input is up.
−Vout is tied to GND here: the robot has one ground, P−. RS+ and RS− tied at the pins, Trim open. Input caps at the pins; the EMC filter is left out.
150 W at 12 V, 89 %: 18 W at full load, about 6 W at 50 W. Natural-air derating starts near 30 C. 12.7 mm tall: keep clear of the fan gap.
INA228 addresses: L arm 0x40, R arm 0x41, waist 0x42, aux 0x43, L leg 0x44, R leg 0x45, 12 V 0x46. Powered by the Pi's 3.3 V; Pi pull-ups on I2C.

5a. Brick soft start (ahead of the E-stop switch)


RS9
BAT
SH9_P
SH9_N
5mR CSS4J-4026K-5L00F
Q7 IPB017N10N5LFATMA1
BRK_G1
BRK_IN
BRK_D
U7 LM5069MMX-2
BRK_PG
BRK_P
BRK_T
R25 10R
R26 37.4k 1%
C19 47nF C0G
C20 100nF 100V
TP5
TP BRK_IN

Same controller as the bank, so one part family. −2 retries on its own: a compute fault does not need a BMS cycle. UVLO on VIN, OVLO low: on whenever the pack is.
Power-limit start, no dV/dt network: 60 W limit (Rs 5 mOhm, 5 mV floor at 54.6 V), about 1.1 A at full pack. 4.4 uF plus the brick's own input caps (not published) charges in under 1 ms at 20 uF.
Timer 47 nF: 2.2 ms typ, 1.5–3.8 ms. Limit 9.7–12.3 A clears the worst brick draw, 4.7 A at 36 V. Fault check: 75 W is 1.4 A for up to 3.8 ms, against 12 A at 10 ms on the datasheet SOA. Q7 is the linear FET for that reason.
Compute is on whenever the BMS is on. Loss at 3 A: 0.05 W shunt, 0.03 W FET.

6. Limb outputs: terminal pair, 1 mOhm shunt, INA228. Fuses inline in the pigtails.


RS2
RS3
RS4
RS5
RS6
RS7
1mR CSS4J-4026K-1L00F 6W
VBUS
SHL1_P
SHL1_N
U11
U12
U13
U14
U15
U16
INA228AIDGSR
+3V3_PI
I2C_SDA
I2C_SCL
C21 100nF
C22 100nF
C23 100nF
C24 100nF
C25 100nF
C26 100nF
V_L1
V_L2
V_L3
V_L4
V_L5
V_L6
J8
J9
J10
J11
J12
J13
J14
J15
J16
J17
J18
J19
L arm +
L arm −
R arm +
R arm −
waist +
waist −
aux +
aux −
L leg +
L leg −
R leg +
R leg −

1 mOhm: 60 A peak is 60 mV, ADCRANGE 0. 0.36 W at 19 A.
Legs DNP: fit the same parts when they arrive.

Rev B: a part goes in only if leaving it out would hurt. Parts list: hardware/electrical/pdb/REV_B_PAR
marengo
Sheet: /
File: marengo_pdu_revb.kicad_sch
Title: marengo power board rev B
Size: A1
Date: 2026-11-06
Rev: B-wip
KiCad E.D.A. 10.0.8
Id: 1/1